

Game Engine Concepts

Fachhochschule Oberösterreich

Human Interface Devices

Bad Game Interface Aspects:

Half-Life 2: Important tasks explained only once (via audio)



Bad Game Interface Aspects:

Skyrim: Ambiguous bag – what am I wearing, and how to compare items?

ALL

APPAREL

POTIONS

BOOKS

MISC

◇ College Robes

◇ Gloves

Lockpick (15)

◇ Mage Hood

Potion of Minor Healing (3)

Potion of Minor Stamina (5)

The Book of the Dragonborn



COLLEGE ROBES

ARMOR 0

WEIGHT 1

VALUE 10

[E] Equip [R] Drop [F] Favorite

Armor Rating 0 Carry Weight 11/300 Gold 34

Bad Game Interface Aspects:
Fallout 4: Why so small?



[A] TOGGLE ACTIVE [X] SHOW ON MAP [R] NAVIGATE OBJECTIVES [Y] PERK CHART



Bad Game Interface Aspects:

GTA San Andreas: Why do NPCs have different walking speeds than I have?



Bad Game Interface Aspects:

Resident Evil 2: Who can move correctly with these ill-placed cameras?

Human Interface Devices

Bad Examples

→ NES Classic Controller: Ergonomics, which hands would fit this?



Human Interface Devices

The Importance of Human-Machine Interfaces

“Better” interfaces guide the evolution of computing

Examples

➔ From **command-line** interfaces...

```

40 GETA$: IFA$(<)CHR$(13) THEN 40
42 PL=PL+1: IF PL>PY THEN PL=1
45 B$=""
    " : V$=""
50 TS=TI: TX=TS+6000: PC=1: PK=33268: D=1:
EM ** DRAW BORDER *
60 PRINT "L" B$ "S" PLAYER"PL" - "LE
T$(PN$(PL), 10) SPC(12);
70 FOR I=1 TO 22: PRINT "R"
75 NEXT: PRINT "R" PRESS ! TO QUIT --
DIFFICULTY = "(20-N%(PL))/2" "LEFT$(B$, 4
80 T=INT(RND(1)*900+32768): IF (33248<T
ND(T<33288) THEN 80
82 TD=1: NN=N%(PL)
85 IF PEEK(T)<>32 THEN 80
90 POKET, 87: REM *** BEGIN ***
99 REM
100 FOR I=1 TO NN: IF PEEK(PE)=81 THEN POKET
BREAK
READY.

```

Human Interface Devices

Examples

→ From **command-line** interfaces...

→ ...to the “**Desktop Metaphor**”



Human Interface Devices

Examples

→ From **Personal Assistants...**



Human Interface Devices

Examples

→ From **Personal Assistants...**

→ To the **Smartphone**



Human Interface Devices

Examples

→ From messengers, blogs, and forums...

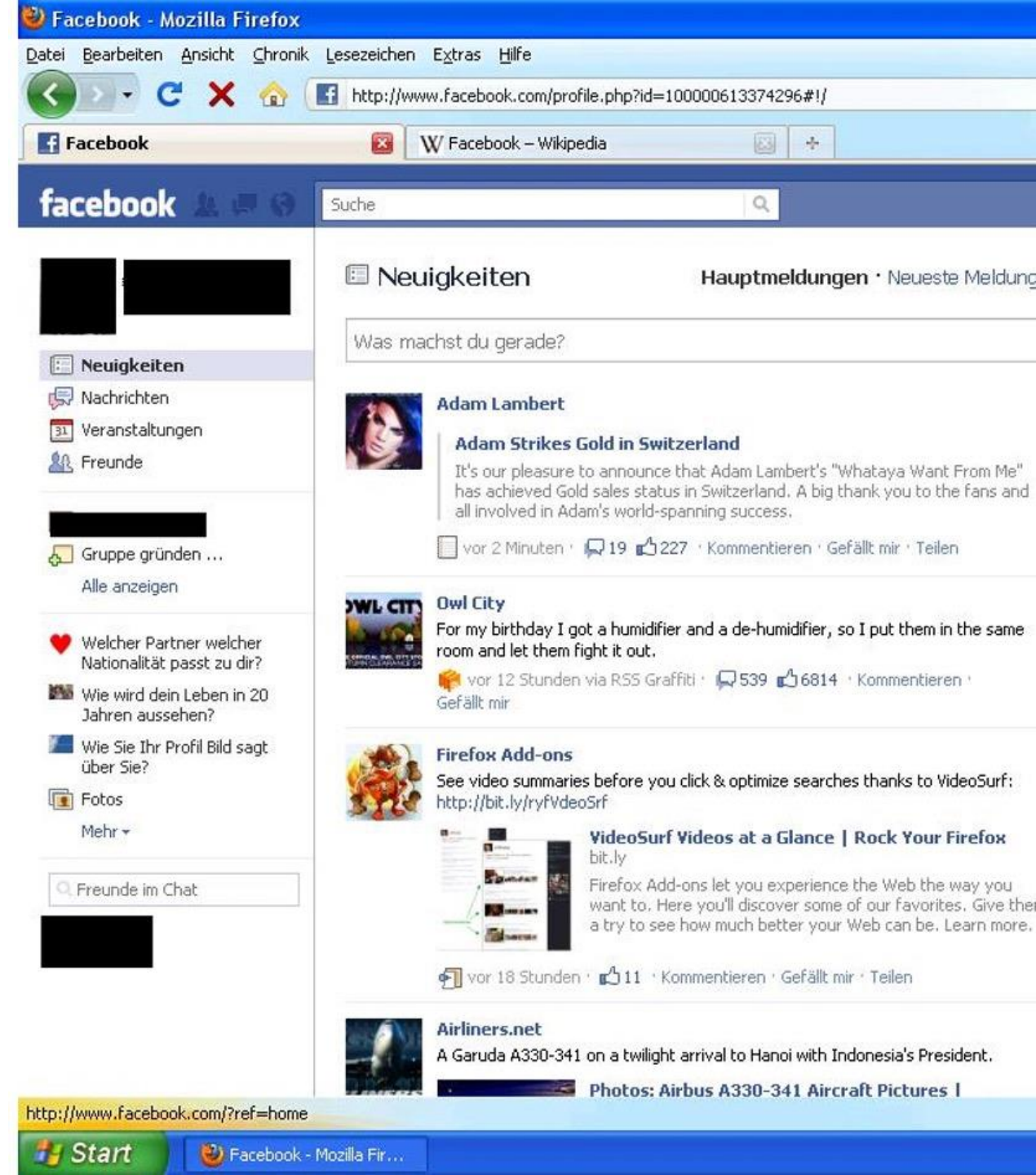


Human Interface Devices

Examples

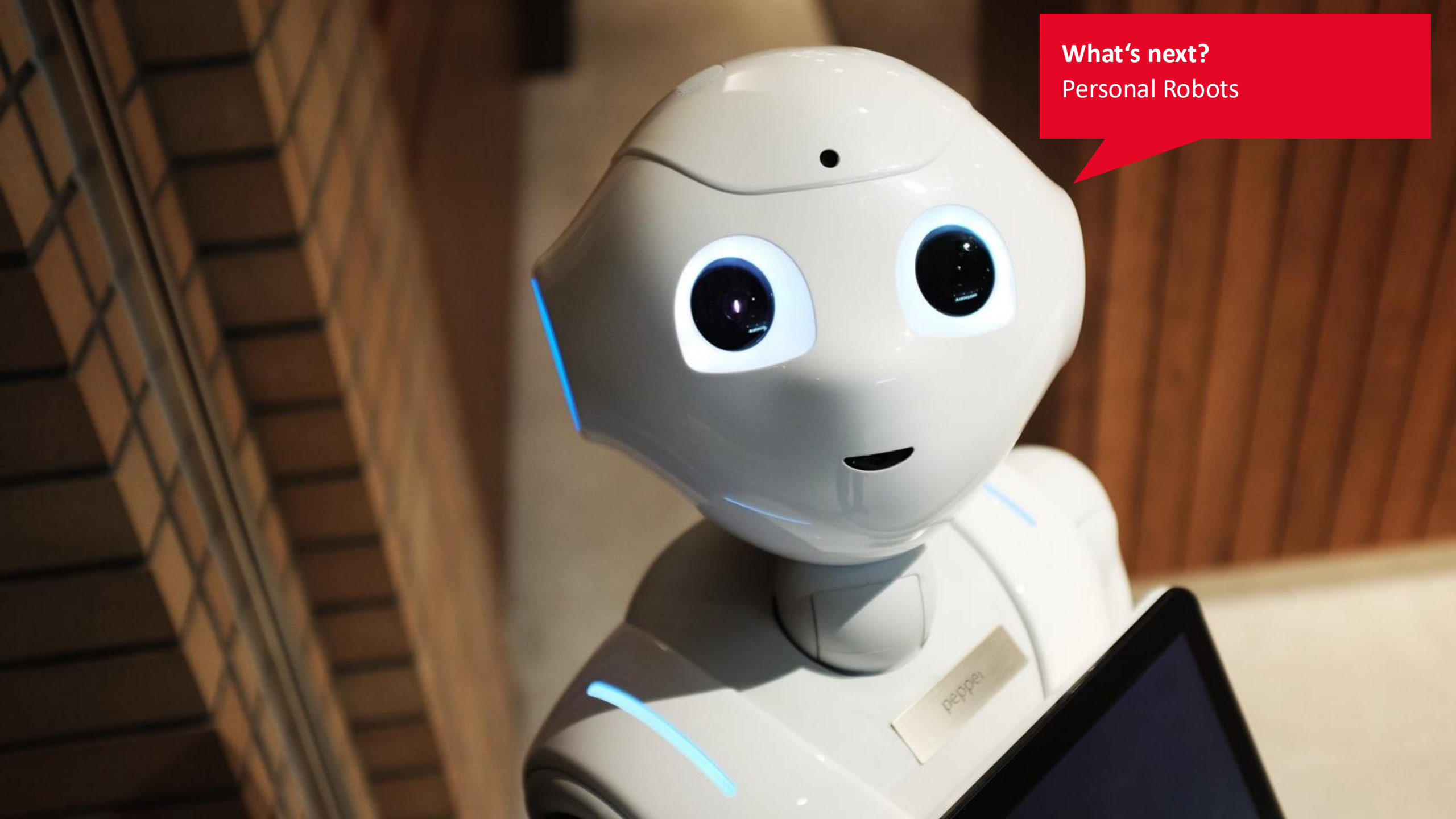
→ From messengers, blogs, and forums...

→ To Social Media





What's next?
Virtual Reality

A close-up, low-angle shot of the Pepper robot, a white, humanoid robot with large, expressive blue eyes and a small black dot for a nose. It is holding a tablet in its right hand. The background is a blurred wooden wall. A red speech bubble is in the top right corner.

What's next?
Personal Robots

What's next?
Automated Vehicles





Human Interface Devices

User Interface (UI)

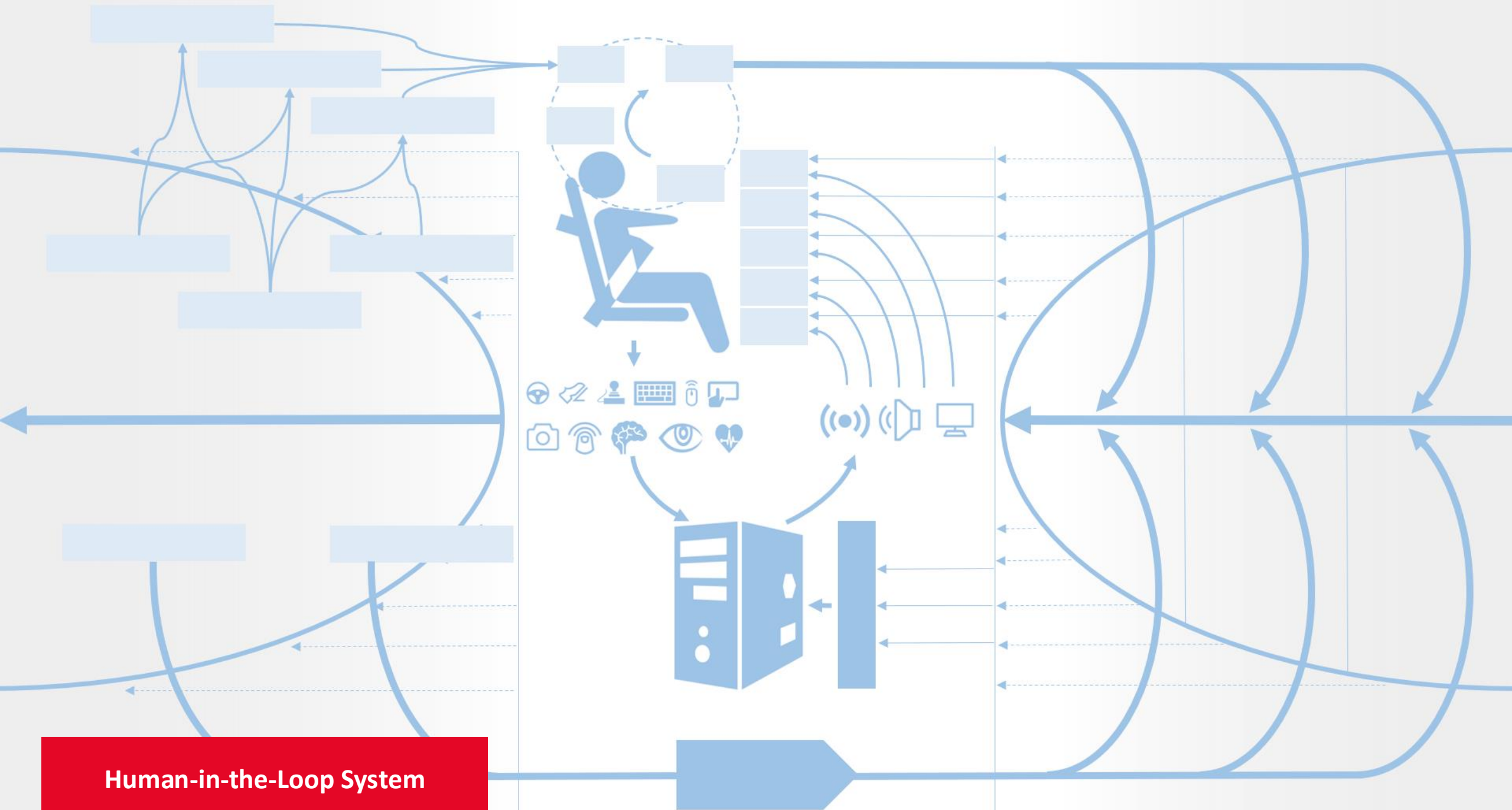
The interface between the user and the information has become the most critical issue for creating effective, efficient and pleasant systems

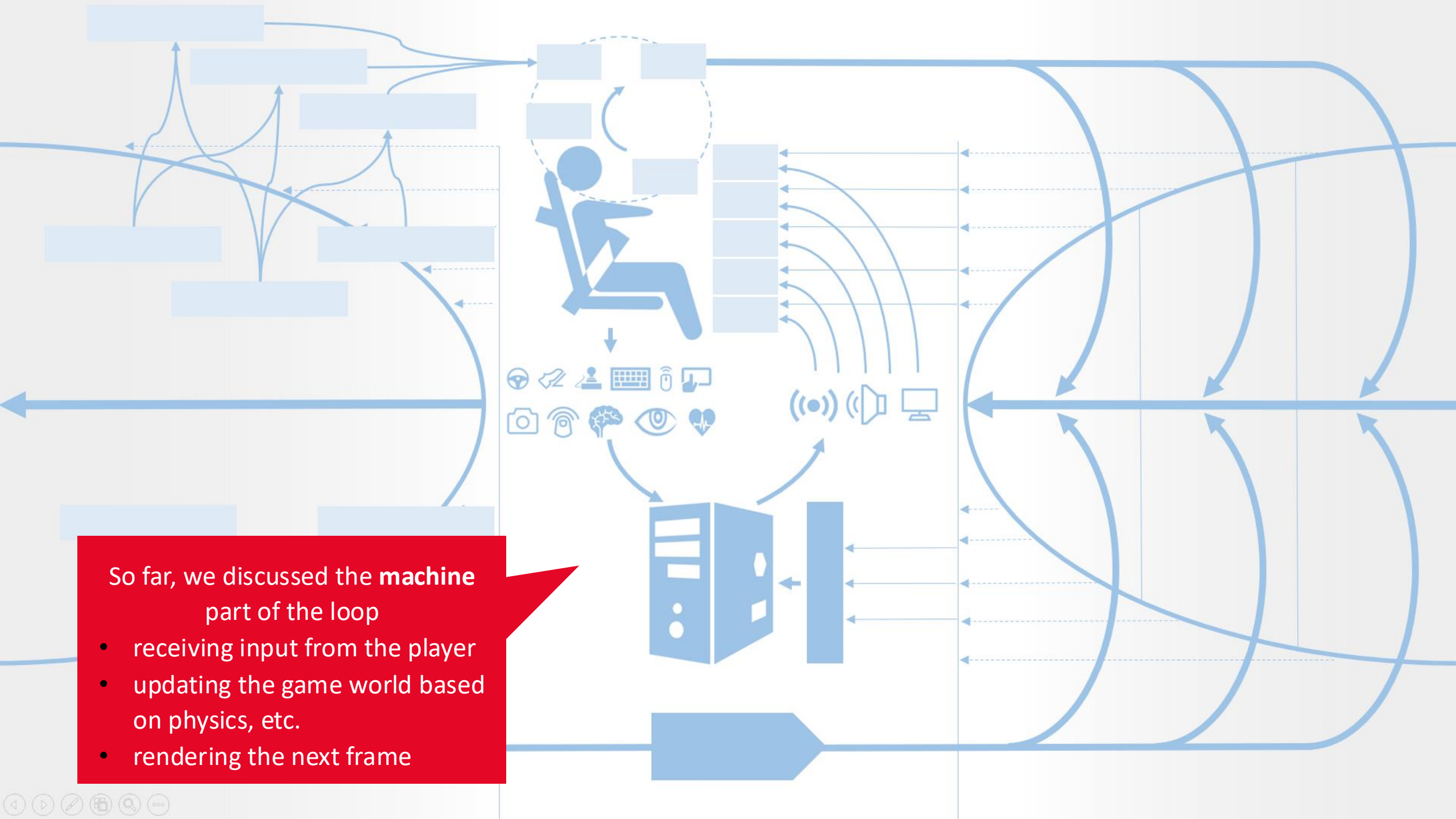


Interaction Design:

Allocating functions to humans and/or technology
Involves sequence/structure of touchpoints of space and time

Benyon, D. (2019). Designing User Experience: a guide to HCI, UX and interaction design. Pearson UK

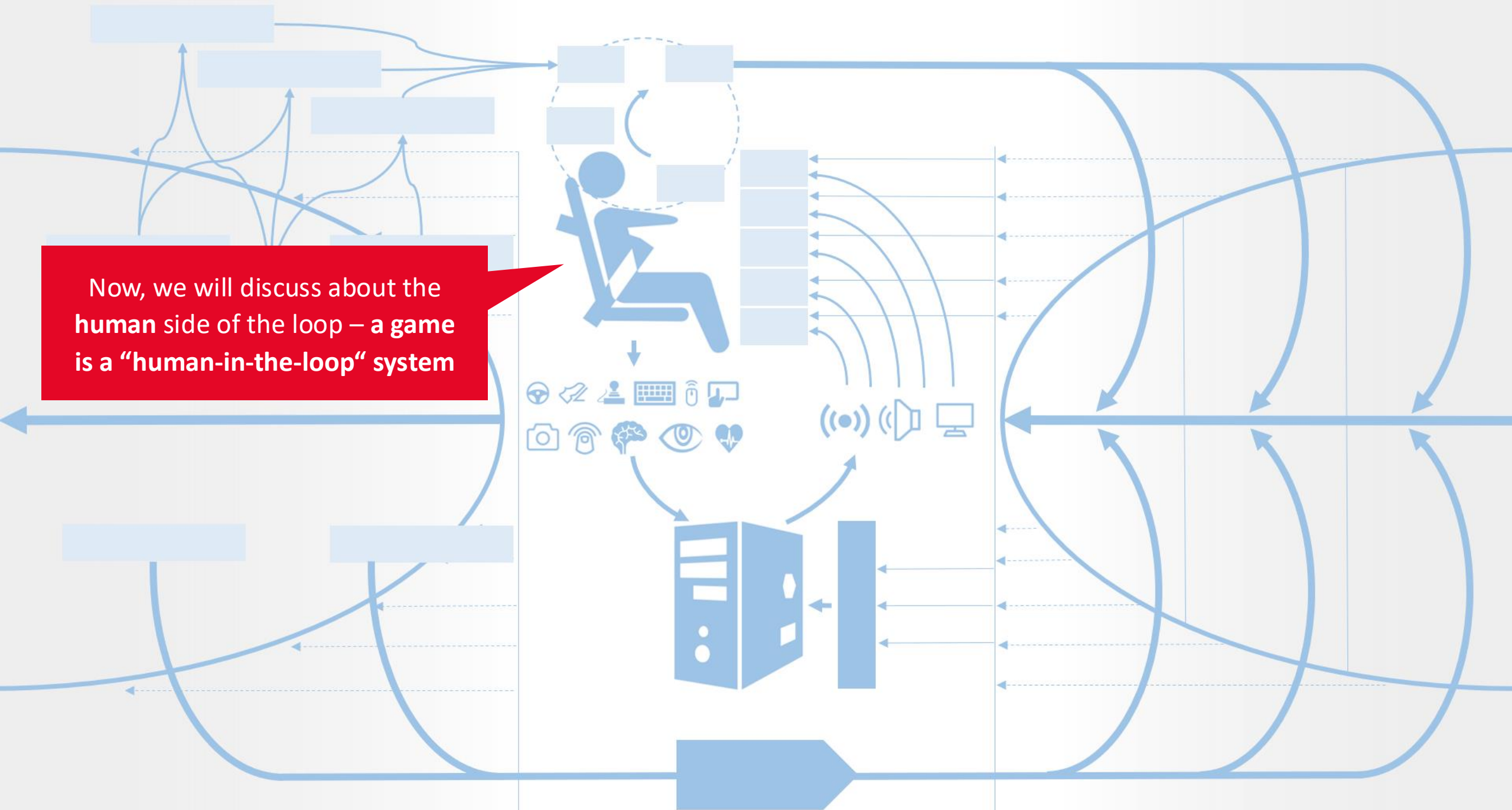




So far, we discussed the **machine** part of the loop

- receiving input from the player
- updating the game world based on physics, etc.
- rendering the next frame

Now, we will discuss about the human side of the loop – a game is a “human-in-the-loop” system



Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario



Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) Output provided by the machine

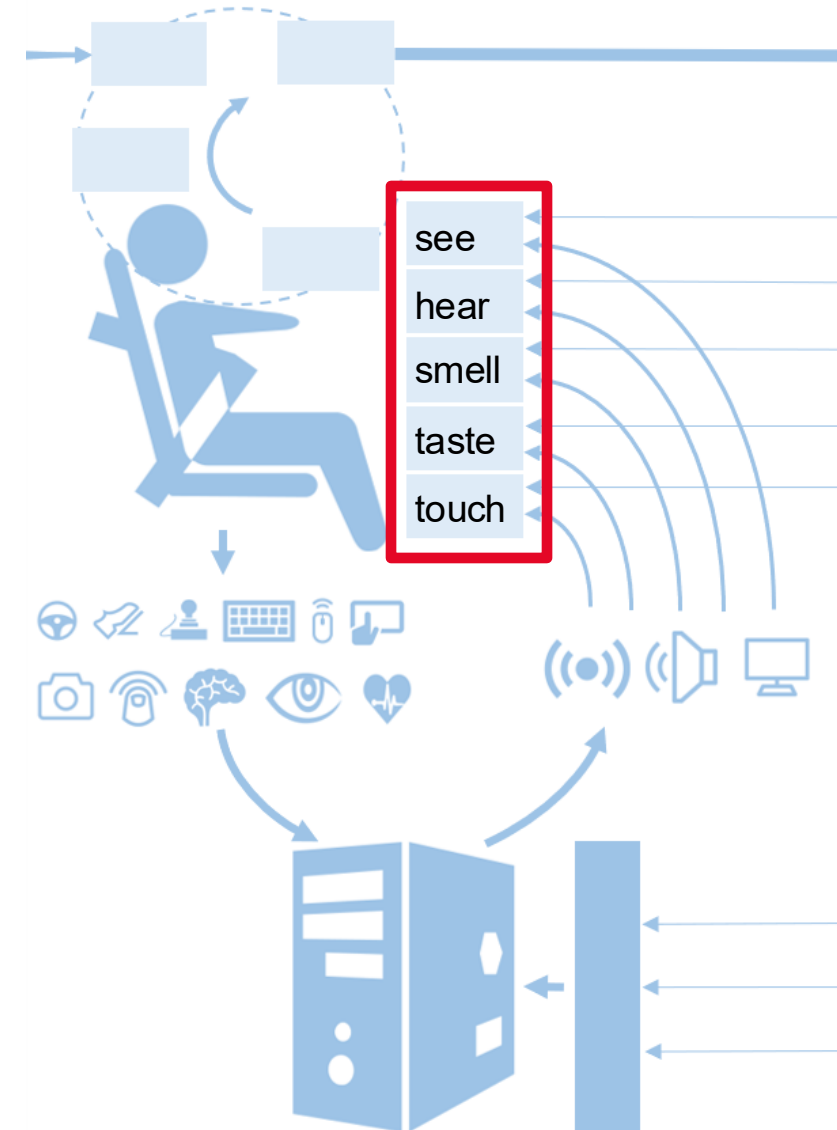


Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
- 2) Human **perception**

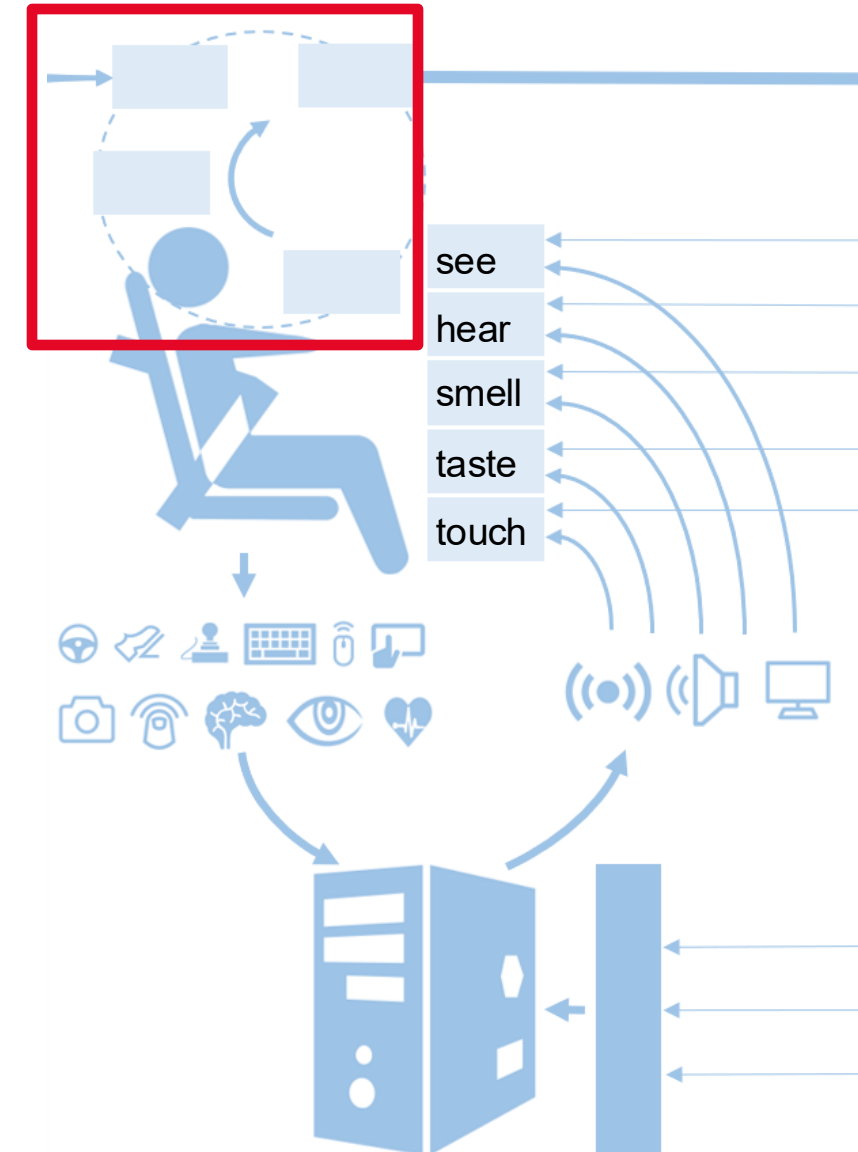


Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
- 2) Human **perception**
- 3) Human processing / decision making

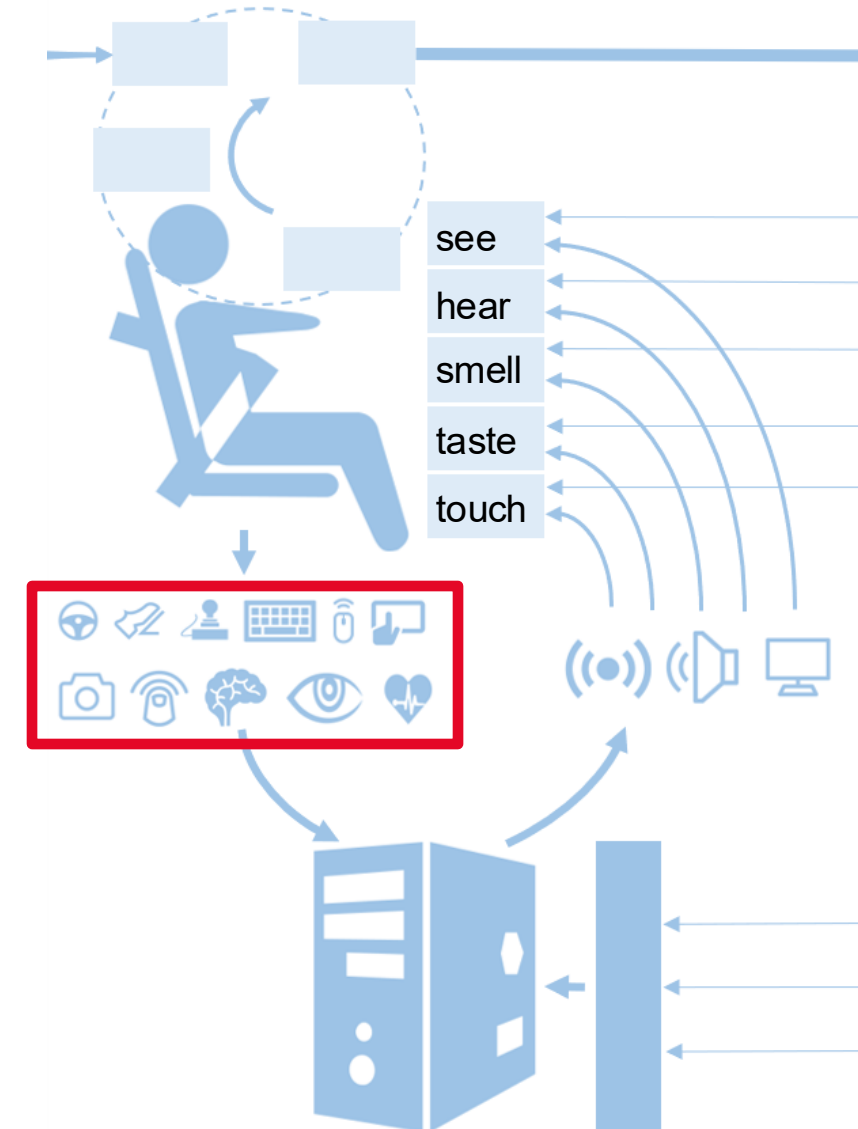


Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
- 2) Human **perception**
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- 4) Input by the human

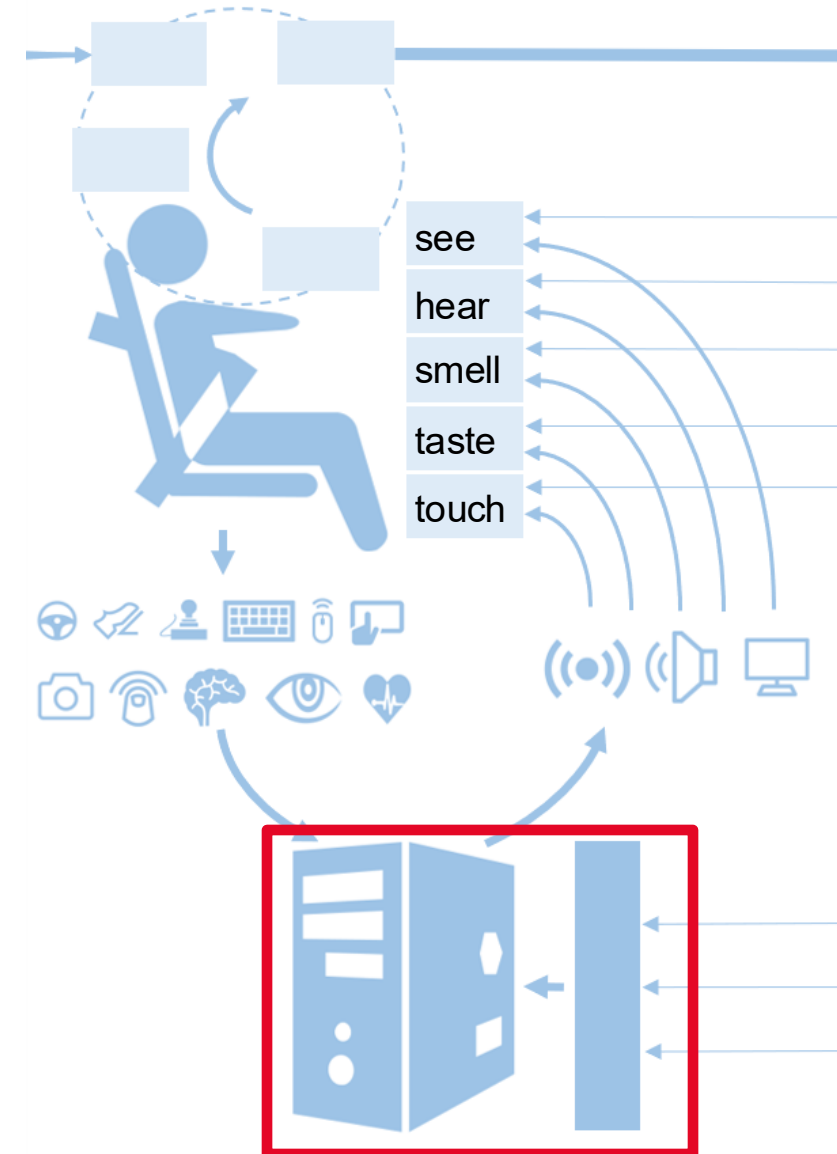


Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
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- 4) Input by the human
- 5) Computation



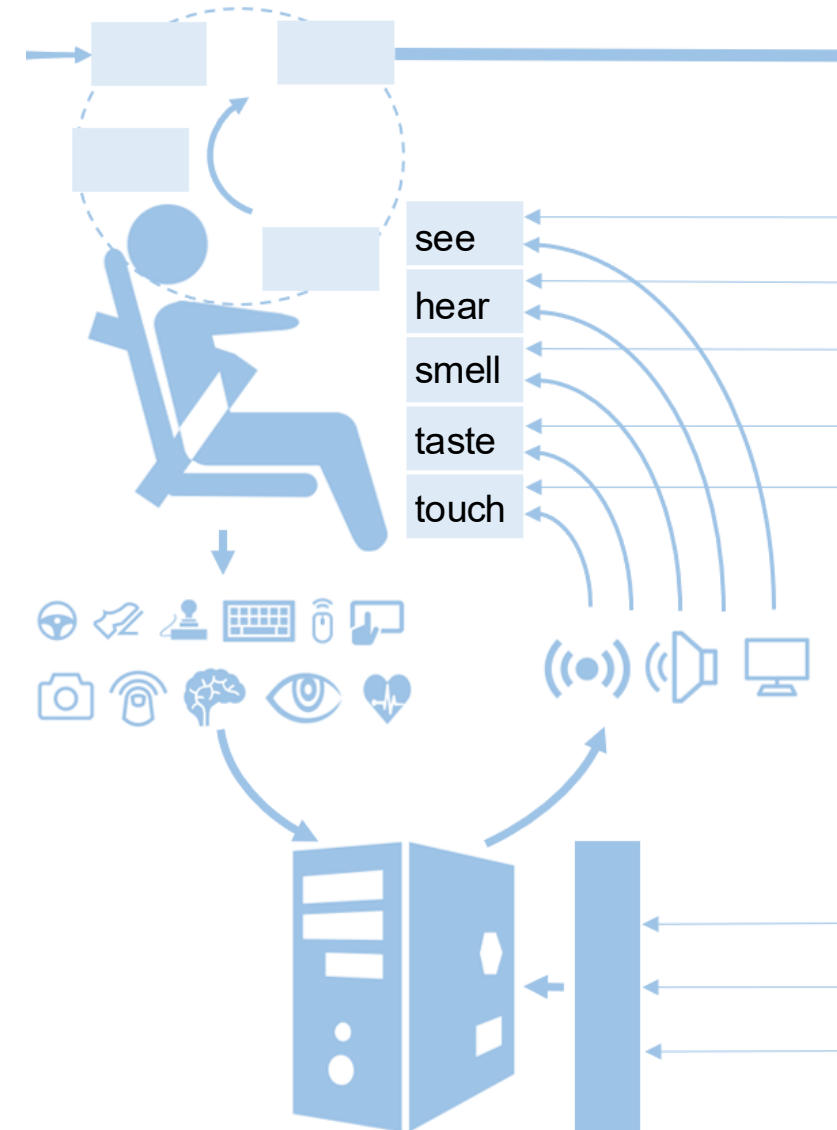
Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
- 2) Human **perception**
- 3) Human processing / decision making
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→ By **HMI Design**, we can modify parts in the loop and evaluate changes in utility/performance



Human Interface Devices

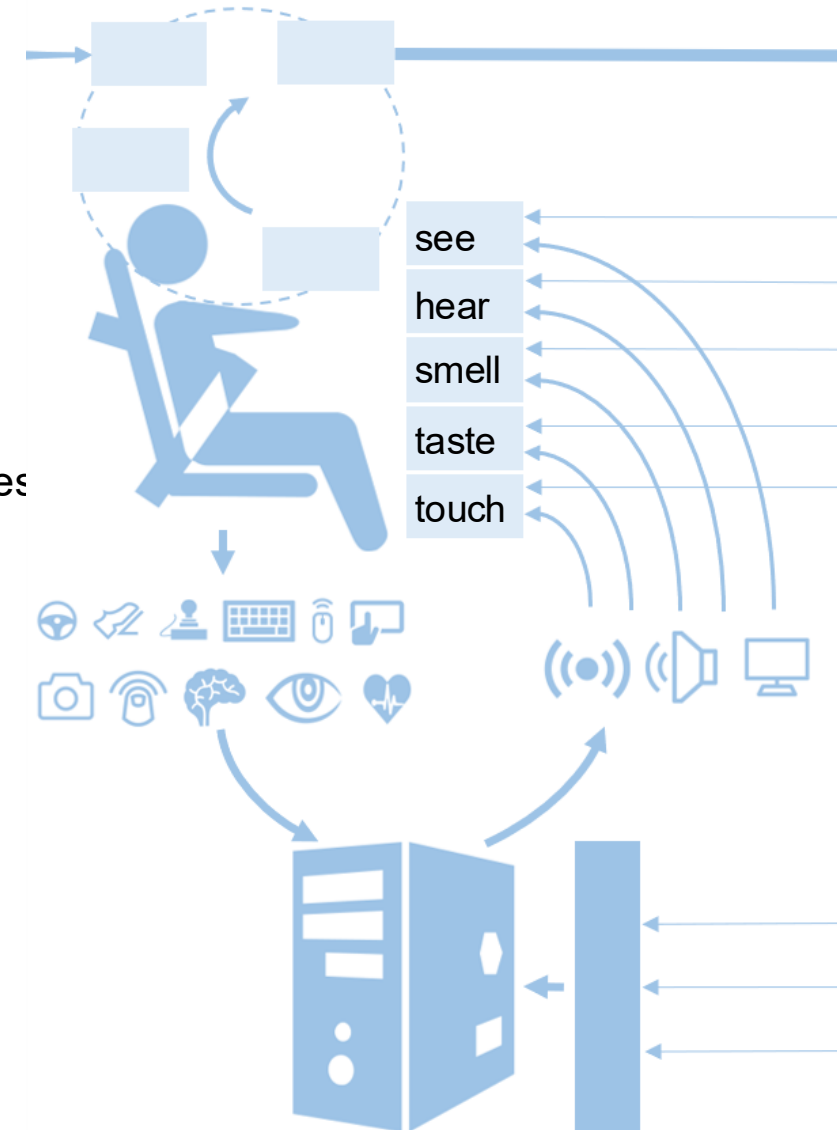
Control Loops

→ Utility/Performance

- There are different optimizations possible
 - Interaction Performance
 - User Experience/Acceptance

→ Simple Example

- Reaction time test – a test subject must press a button as soon as he/she perceives
- How long does this take?
- $RT \text{ (reaction time)} = \text{perception time} + \text{response action time}$
- <https://www.humanbenchmark.com/tests/reactiontime>

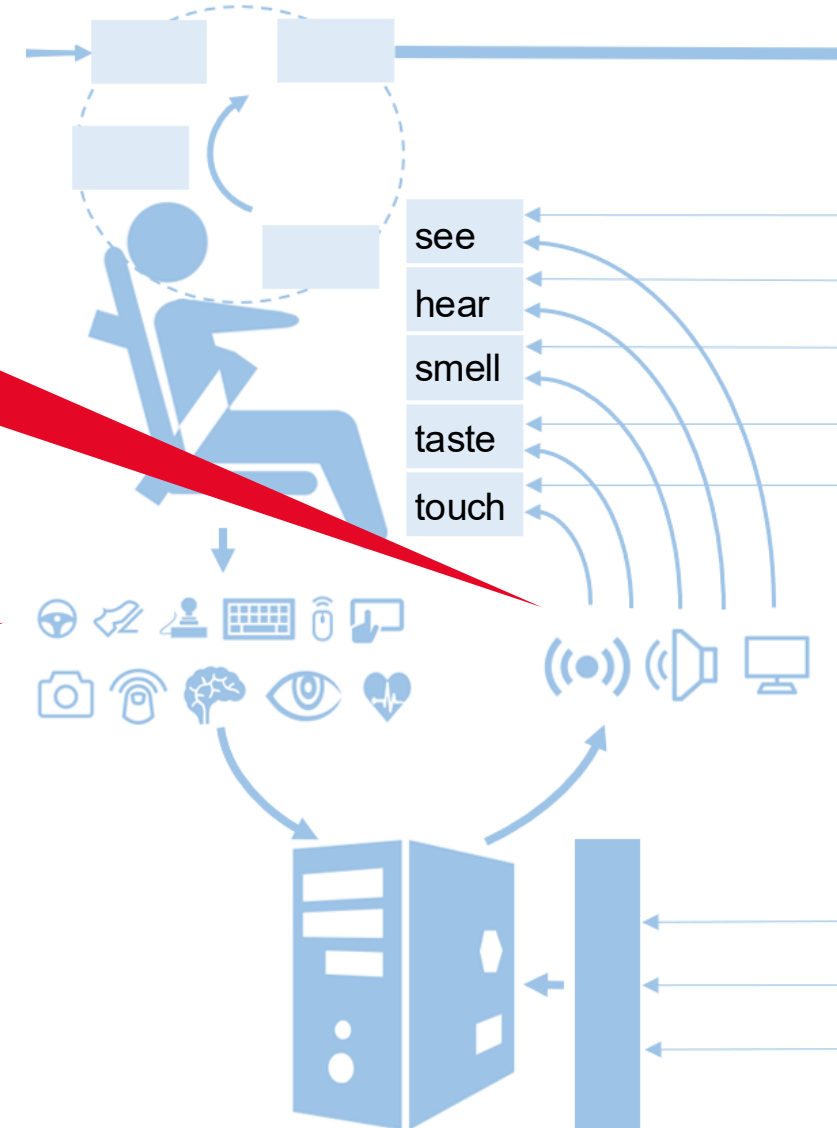


Human Interface Devices

Control Loops

Traditional Interface Perspective:
The user sees the real world and
some machine interface

Traditional Interface Perspective:
Make the human part of the loop as
fast, easy, acceptable, etc. as possible

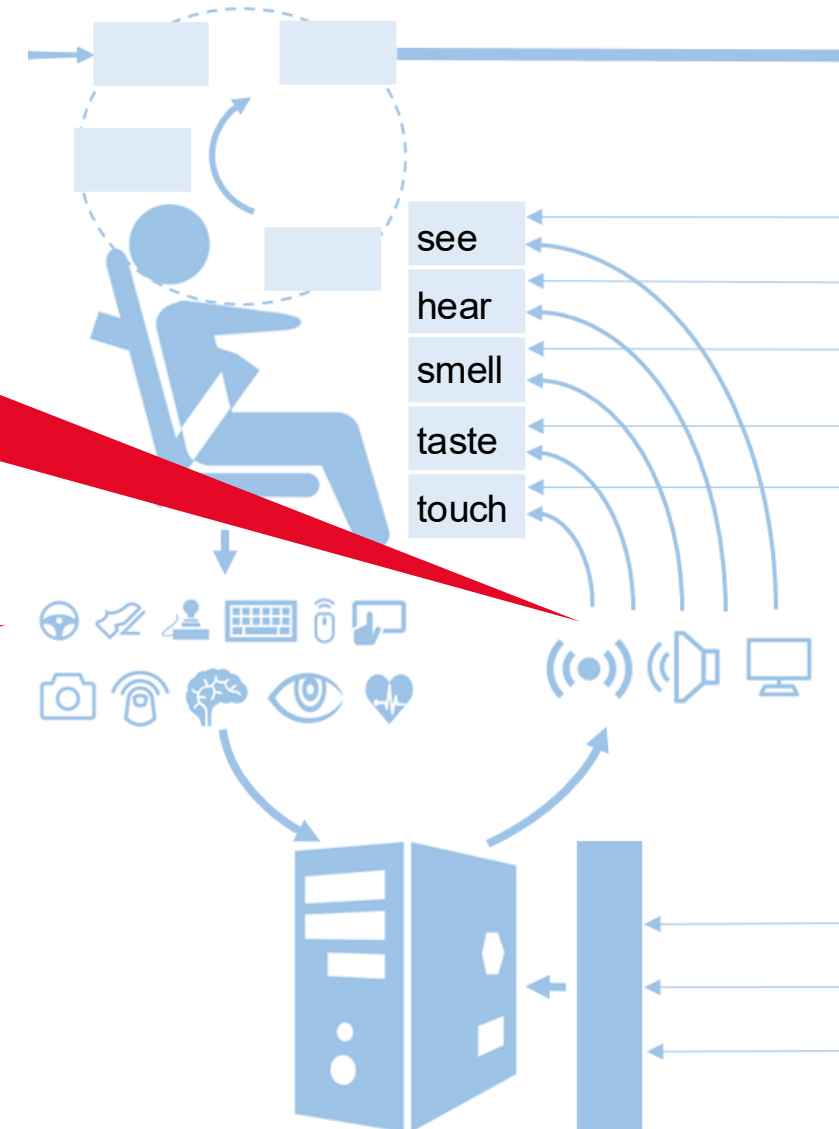


Human Interface Devices

Control Loops

Game Interface Perspective:
There is no “real world” (except for MR),
the virtual world is now a controllable
element (and thus, part of the interface)!

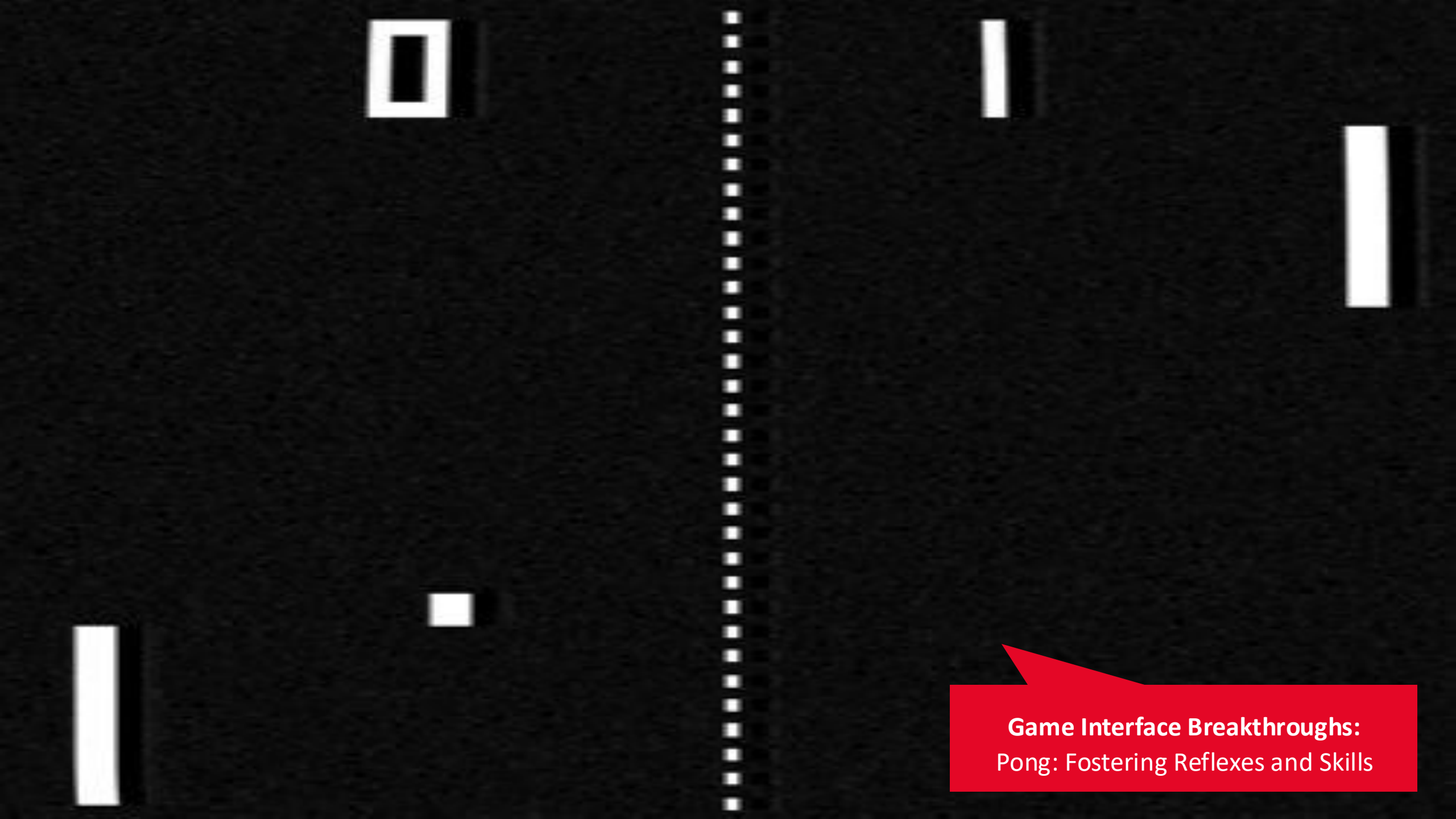
Game Interface Perspective:
Make the human part of the loop as
FUN as possible



A game interface “is the part of the game that allows the user to interact with the game. [It] is the connection between the user and the game”

“the infinitely thin membrane that separates the player
and the game”

It involves input & feedback!



Game Interface Breakthroughs:
Pong: Fostering Reflexes and Skills

1UP
220

HIGH SCORE
1000

2UP
290



Game Interface Breakthroughs:
PacMan: Scores and Puzzle-Solving Activities

Introduction

Somewhere nearby is colossal cave, where others have found fortunes in treasure and gold, though it is rumoured that some who enter are never seen again. Magic is said to work in the cave. I will be your eyes and hands. Direct me with commands of 1 or 2 words. I should warn you that I look at only the first four letters of each word, so you'll have to enter "NORTHEAST" as "NE" to distinguish it from "NORTH". (Should you get stuck, type "HELP" for some general hints. For information on how to end your adventure, etc., type "INFO".)

This program was originally developed by Willie Crowther. Most of the features of the current program were added by Don Woods. The current version was done by Bob Supnik. This version was implemented on the IBM-PC (and compatibles) by Kevin Black.

For further information consult your scroll (READ.ME).

GOOD LUCK!

You are standing at the end of a road be
Around you is a forest. A small stream
down a gully. In the distance there is a

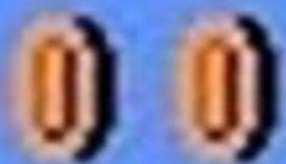
Game Interface Breakthroughs:
Colossal Cave Adventure: Integration of Storylines

MARIO
035350

● × 18

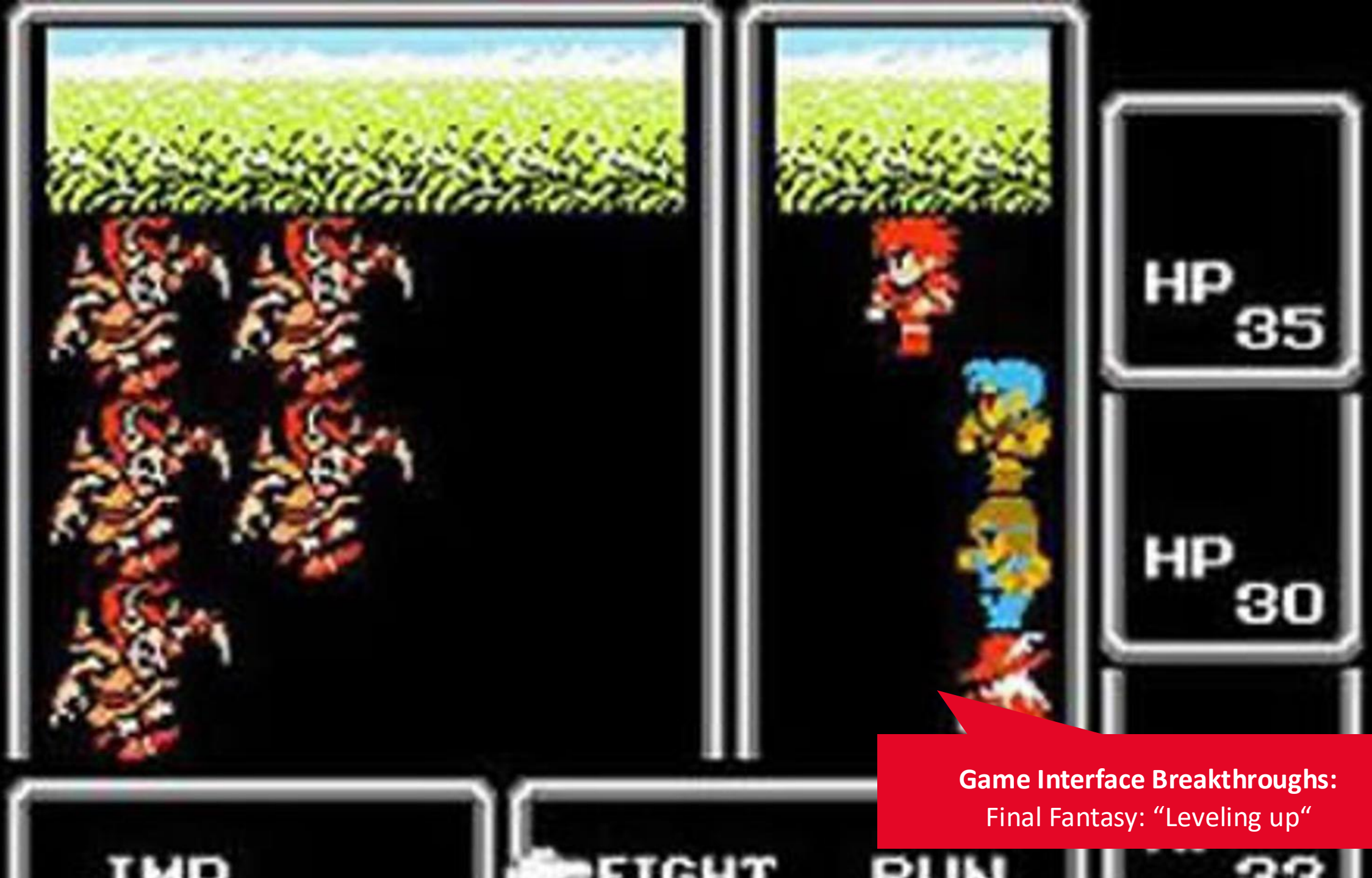
WORLD
1-3

TIME
244



Game Interface Breakthroughs:

Super Mario Bros: Side-Scrolling instead of Page-Flipping



Game Interface Breakthroughs:
Final Fantasy: "Leveling up"



Game Interface Breakthroughs:
Tomb Raider: Action adventures; from a game to a movie



Game Interface Breakthroughs:
Grand Theft Auto: Non-linear gameplay and storylines



Game Interface Breakthroughs:
Crysis: Physics as game element



Game Interface Breakthroughs:
Guitar Freaks: Next generation embodiment



Game Controller Breakthroughs:
Kinect Adventures: Full body movement



Game Controller Breakthroughs:
Missile: Joysticks

Human Interface Devices

Game Interface



Game Controller Breakthroughs:
Nintendo NES Gamepad Controller



Game Controller Breakthroughs:
MacOS Marathon: First Mouse Free-Look

Human Interface Devices

Game Interface



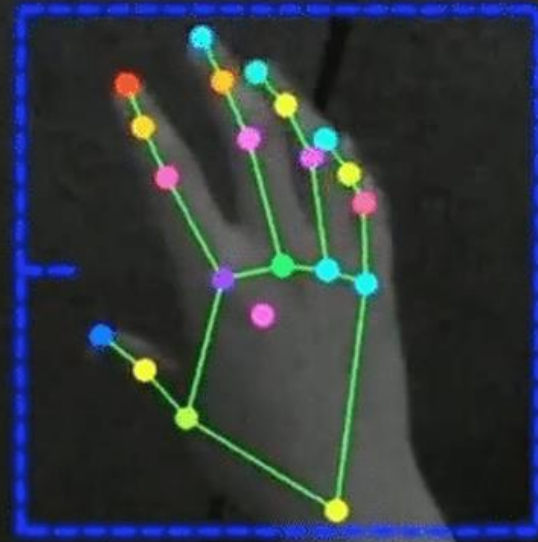
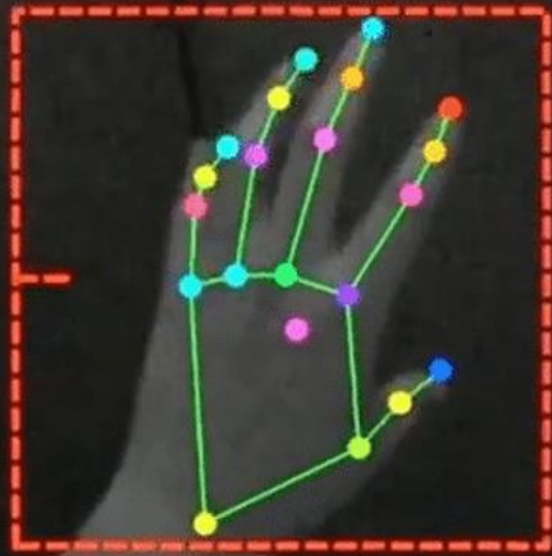
Game Controller Breakthroughs:
Nintendo 64 Analog Stick

Human Interface Devices

Game Interface



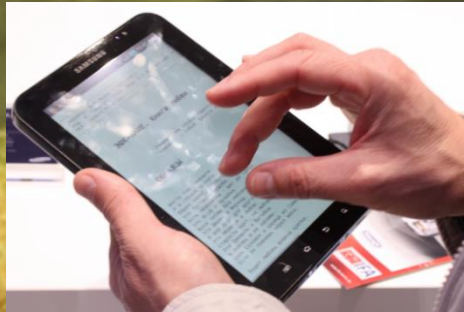
Game Controller Breakthroughs:
Nintendo Wii Nunchuck



Game Controller Breakthroughs:
VR Hand Tracking

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62  



Moorhuhn Jagd (1999)
What is the best input method for this game?

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Solution 1: Experiment

We could invite 20 participants, each of them has to do the task with each approach, we will record how long it takes them, before we conduct a statistical analysis.

Solution 2: Model

We could simulate “artificial users” performing the task. If the simulation is accurate, then a piece of software (or a pen & paper calculation) can tell us which option to choose.

Human Interface Devices

Human Input

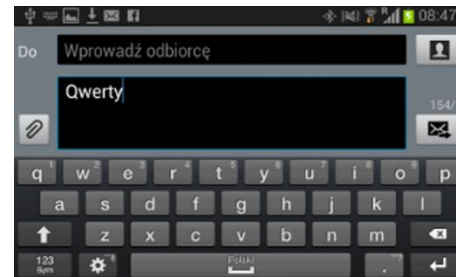
- For both solutions, we need appropriate strategies...
- ... and knowledge about user input!
- Different forms of **(elementary) input tasks**
 - **Text:** specify a series of symbolic data
 - **Position:** specify a position within a range (i.e., screen position with pointing device)
 - **Selection:** choose object from a set of alternatives
 - **Quantify:** specify a numeric value
 - **Orient:** specify an angle or 3-D orientation
 - **Path:** specify a series of positions/orientations

Human Interface Devices

Human Input: Text



Haptic Keyboard



Smartphone Keyboard



Tablet Keyboard



Knob Control



Gamepad



Speech Input

Human Interface Devices

Human Input: Typing Performance Measures

→ Notation

- Presented Text (P): text which has to be typed, $|P|$ is the length of P
- Transcribed Text (T): text entered by user
- Correct (C): number of corrected characters in T

→ Entry Rates

- Completion Time (t): time needed to reproduce text
- Keys per Second (KPS): All keys in IS divided by t
- Characters per Second (CPS): All characters in T divided by t
- Keystrokes per Character (KSPC): Length of IS divided by length of T
- Words per Minute (WPM)

→ Error Rates

- Error Rate: number of incorrect chars divided by $|T|$
- Total Error Rate (Total ER)

Human Interface Devices

Human Input: Typing Performance Measures

→ Average Words per Minute (WPM)

- Physical Keyboard: M=75.85 SD=15.61
- Virtual Keyboard Projection: M=46.60
- Stylus: M=11.62 SD=3.37

→ Average Error Rate (ER)

- Physical Keyboard: M=1.8
- Virtual Keyboard Projection: M=3.7
- Stylus: M=16.475 SD=6.31

→ What about Speech Input?

- Obviously depending on the quality of the speech recognition
- Good performance is possible, but how to treat potential errors?

Human Interface Devices

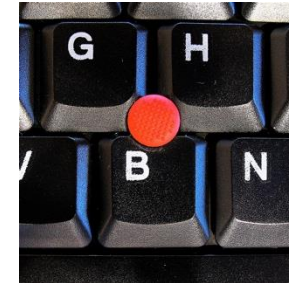
Human Input: Position



Mouse



Trackball



Trackpoint



Joystick



Touch

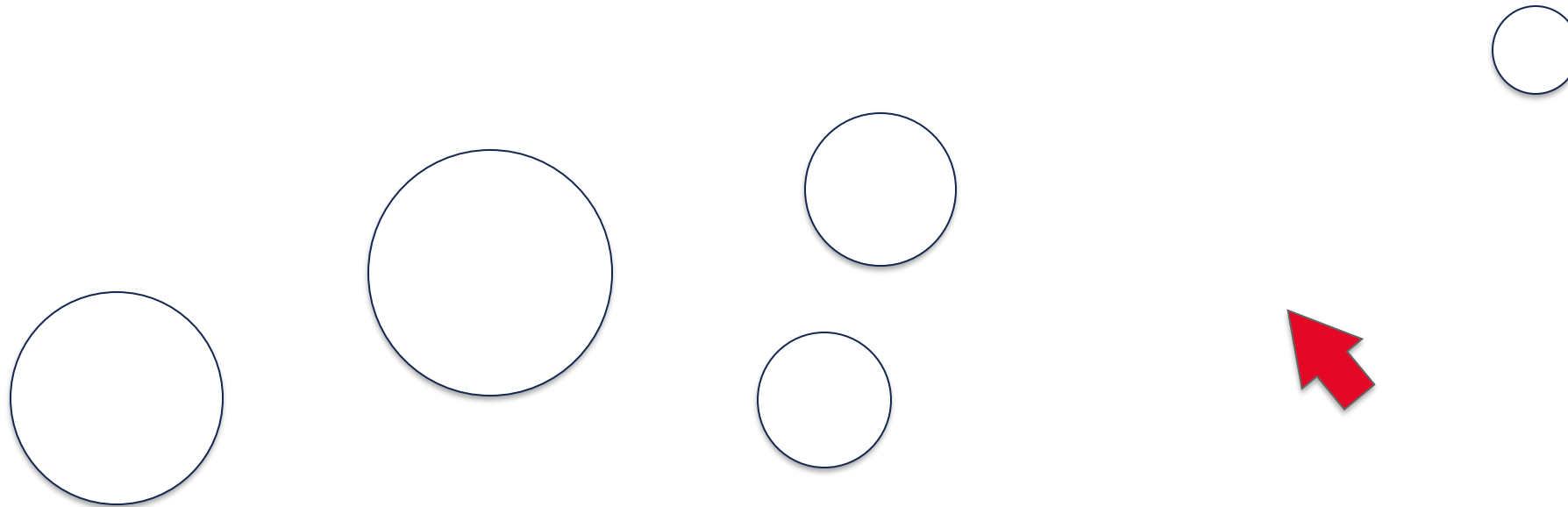


Gestures

Human Interface Devices

Human Input: Position

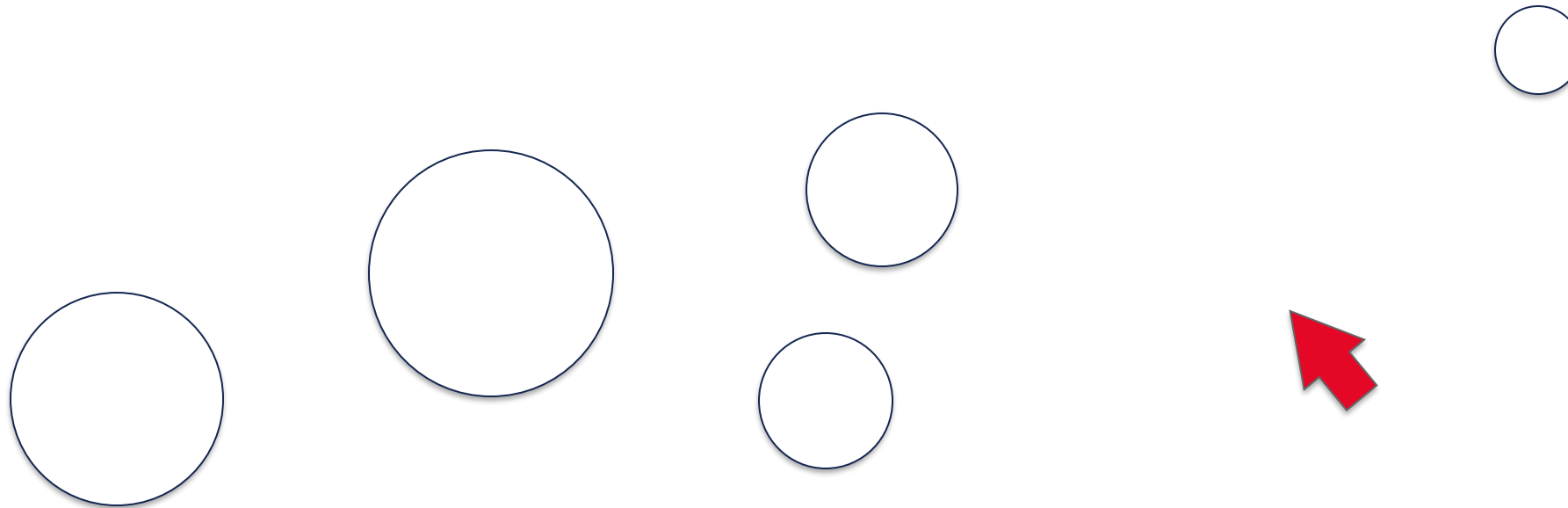
→ **Fitts' Law:** Which targets are easy to reach, and why?



Human Interface Devices

Human Input: Position

→ **Fitts' Law:** Which targets are easy to reach, and why?



→ A **near** target is better than a far target

→ A **large** target is better than a small target

Human Interface Devices

Human Input: Pointing Performance Measures

→ Fitts' Law

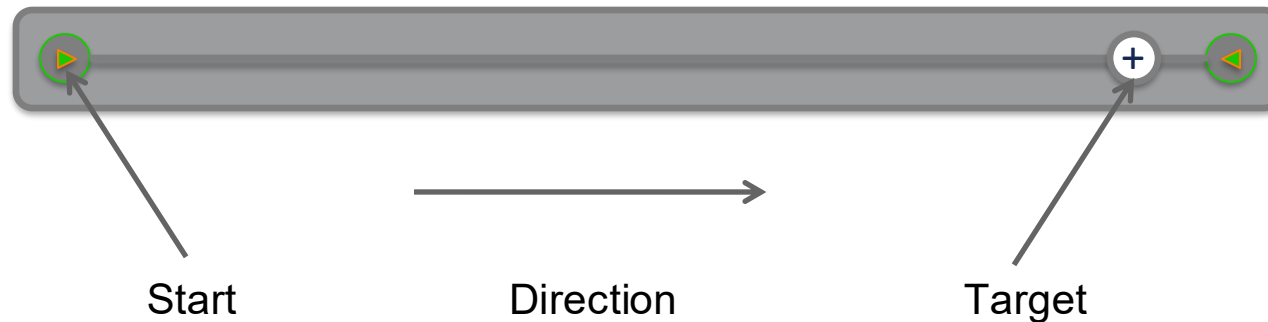
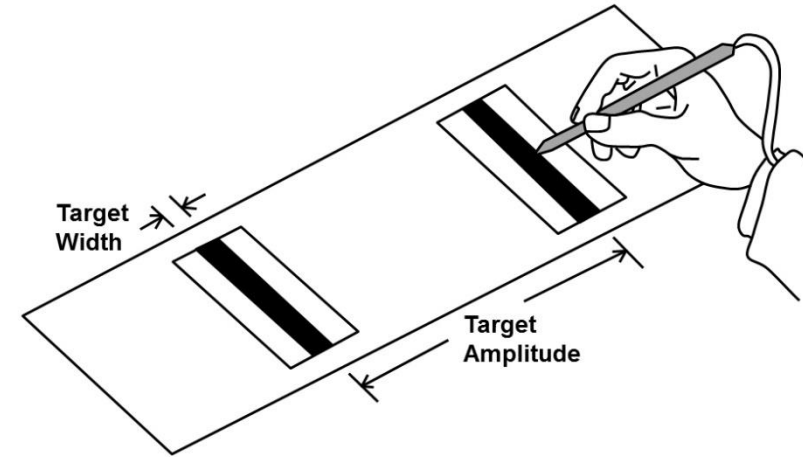
- One of the few “hard facts” in HCI
- Allows to **predict the completion time of pointing tasks** (instead of just implementing and measuring it afterwards)
- Originally developed for 1-dimensional pointing tasks
- Longer distances take longer time
- Larger target areas are easier to “hit”
- The law only **applies to linear movements** (i.e., no cursor acceleration)
- Training can improve performance (as in all tasks)

Human Interface Devices

Human Input: Pointing Performance Measures

→ Fitts' Law in 1-dimensional space

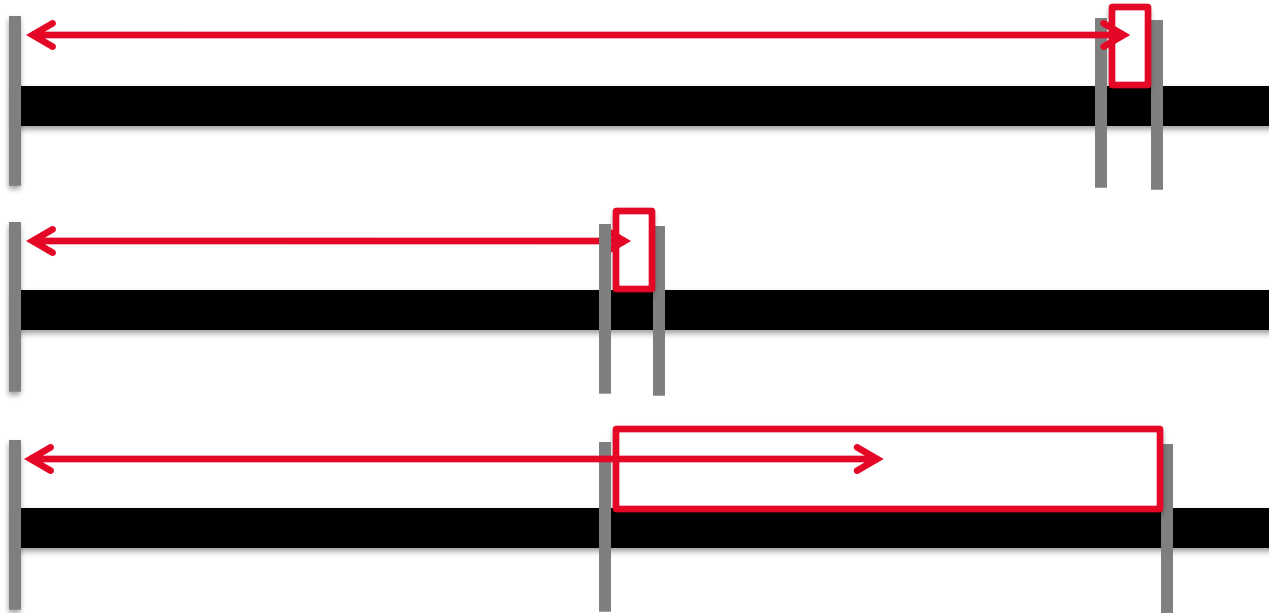
- Original-Experiment Fitts 1954
- Only left-right pointing tasks (“serial tapping task”)



Human Interface Devices

Human Input: Pointing Performance Measures

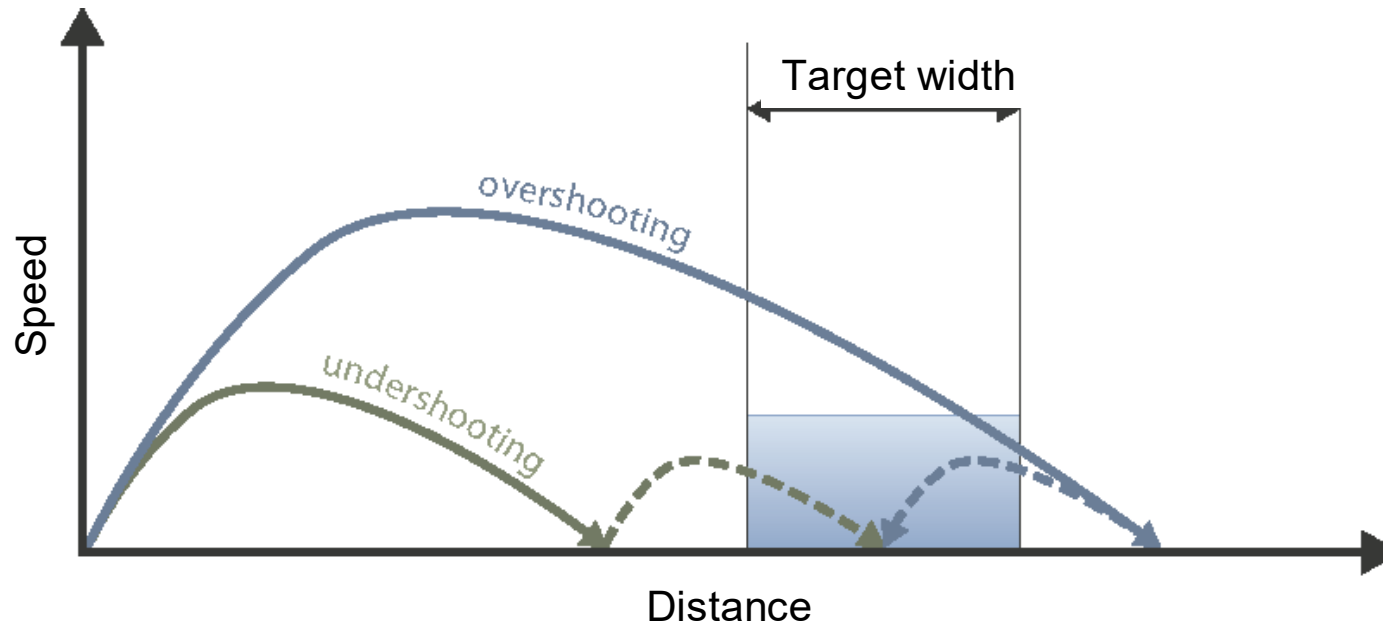
→ Fitts' Law in 1-dimensional space



Human Interface Devices

Human Input: Pointing Performance Measures

→ Fitts' Law in 1-dimensional space

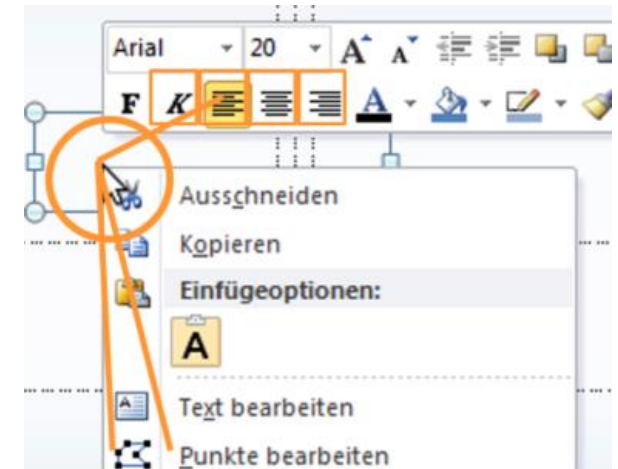
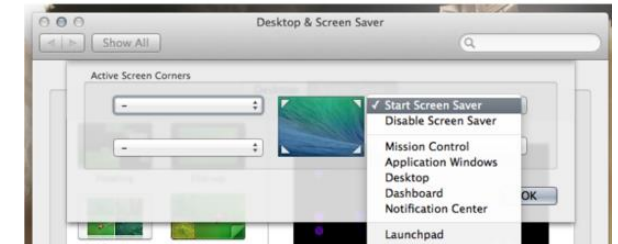
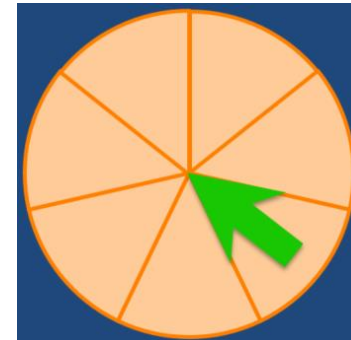


Human Interface Devices

Human Input: Pointing Performance Measures

→ Fitts' Law: Implications for (Game) Interface Designs

- 5 positions most easy to reach
 - 4 corners (i.e., “hot corner” in MS Windows, Mac OS)
 - Current cursor position
- Position, arrangement, and size of context menu items
- Growing targets
- Pie menus, just 1px distance to the target, large buttons



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Moorhuhn Jagd (1999)

We can model, calculate, and predict the performance using Fitts' Law!

Human Interface Devices

Human Input: Selection

→ Choose object from a set of alternatives

- **Cognitively demanding** and difficult, especially if options are similar
- Applies to menus in computer programs, file lists, etc.
- There is no “input device” for selection
- However, **selection has impact on performance** and can be modelled
- **Hicks’ Law models selection tasks** in ordered lists
 - “The time needed to make a selection is proportional to the log number of alternatives given”
 - Hick’s Law does not apply to linear search, i.e., randomly ordered lists
 - Alternatives must be related; this must be obvious to users
 - **Consequence:** menus should be ordered



Human Interface Devices

Human Input: Selection Performance

→ Which list allows more easily to find a certain country?

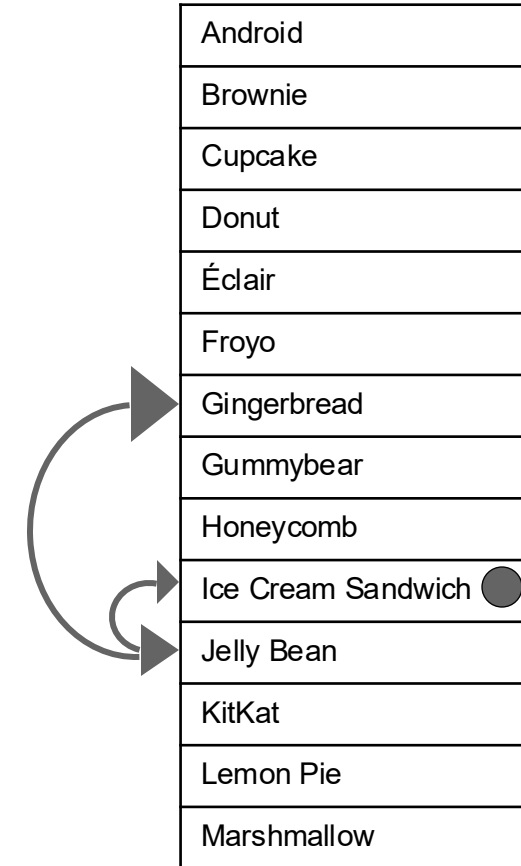
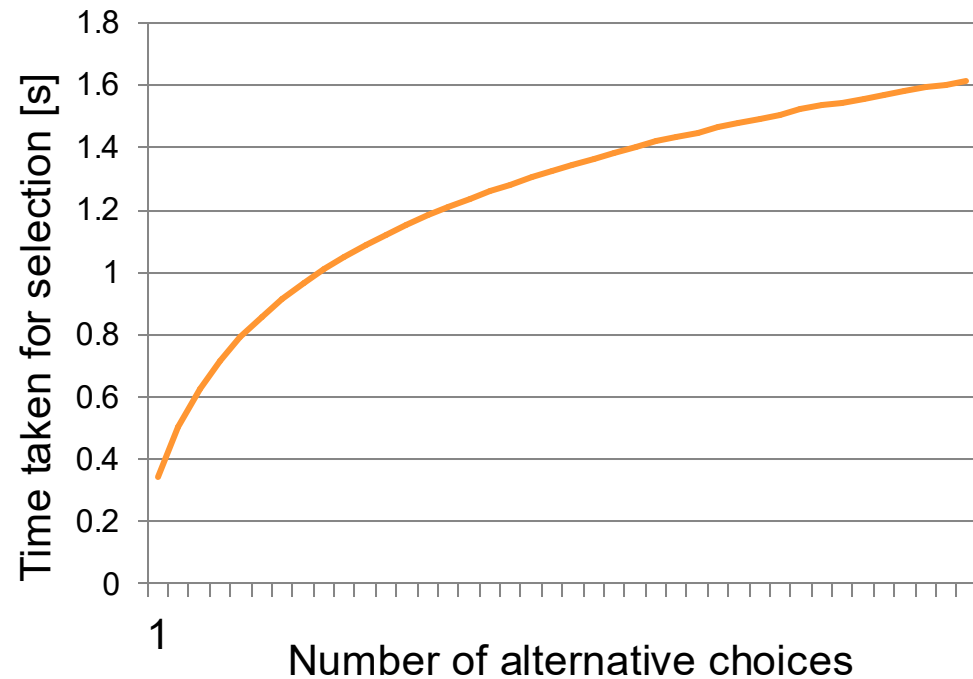
- | | |
|------------------|------------------|
| ○ Austria | □ Luxembourg |
| ○ Belgium | □ Malta |
| ○ Bulgaria | □ Netherlands |
| ○ Cyprus | □ Poland |
| ○ Czech Republic | □ Sweden |
| ○ Denmark | □ United Kingdom |
| ○ Estonia | □ Ireland |
| ○ Finland | □ Hungary |
| ○ France | □ Belgium |
| ○ Germany | □ Finland |
| ○ Greece | □ Czech Republic |
| ○ Hungary | □ France |
| ○ Ireland | □ Germany |
| ○ Italy | □ Greece |
| ○ Latvia | □ Bulgaria |
| ○ Lithuania | □ Cyprus |
| ○ Luxembourg | □ Slovakia |
| ○ Malta | □ Slovenia |
| ○ Netherlands | □ Latvia |
| ○ Poland | □ Portugal |
| ○ Portugal | □ Austria |
| ○ Romania | □ Denmark |
| ○ Slovakia | □ Estonia |
| ○ Slovenia | □ Lithuania |
| ○ Spain | □ Spain |
| ○ Sweden | □ Italy |
| ○ United Kingdom | □ Romania |

Human Interface Devices

Human Input: Selection Performance

→ Hick's Law is similar to Fitts' Law

- The logarithmic nature of the function seems obvious
- In every step, (roughly) half of the options are eliminated

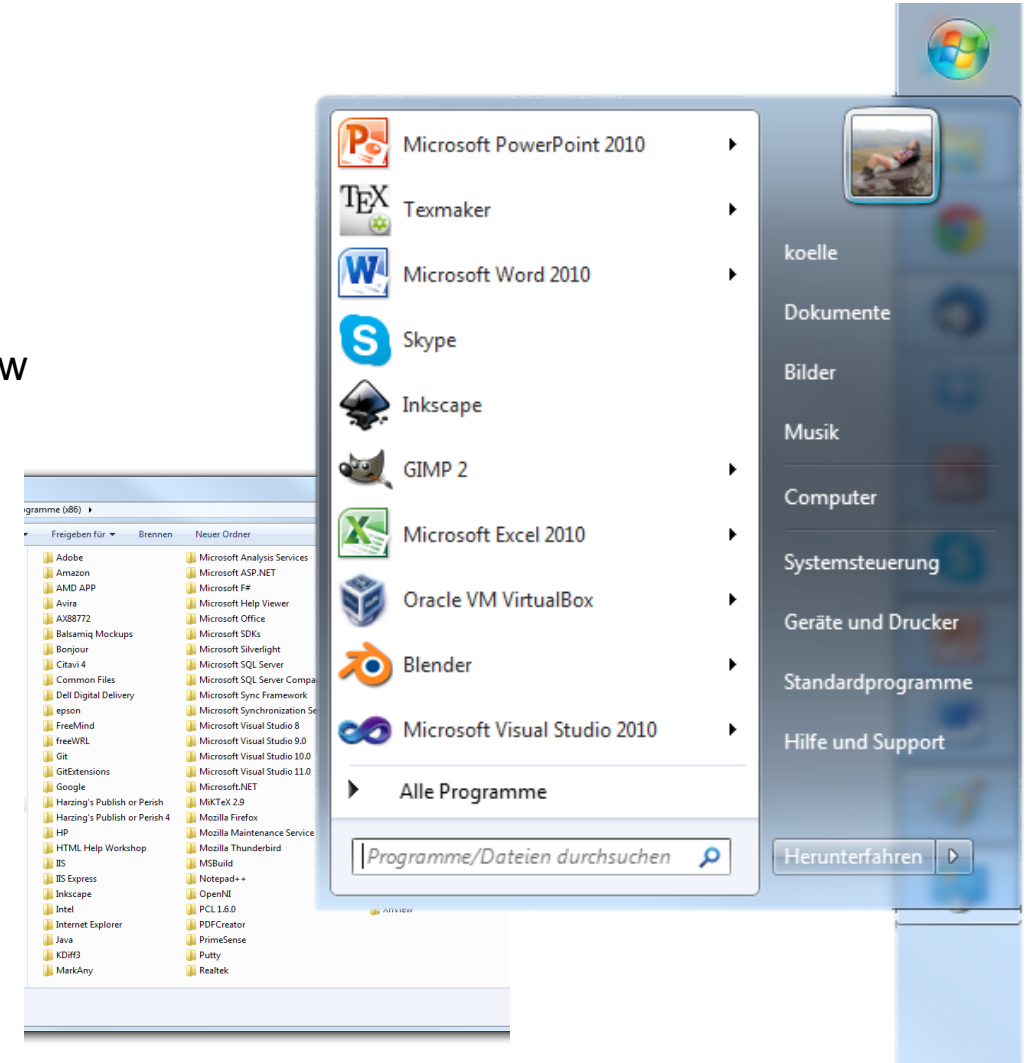


Human Interface Devices

Human Input: Selection Performance

→ Hick's Law Examples

- Used to decide **how menus are arranged**
- If the list is not ordered, search is linear
- Hicks' Law does not apply!
- **Icons make search easier**, but this has nothing to do with Hicks' Law
- If list is large and needs to be displayed in sub-lists beside each other, Hicks' Law does not apply (even when alphabetically sorted)

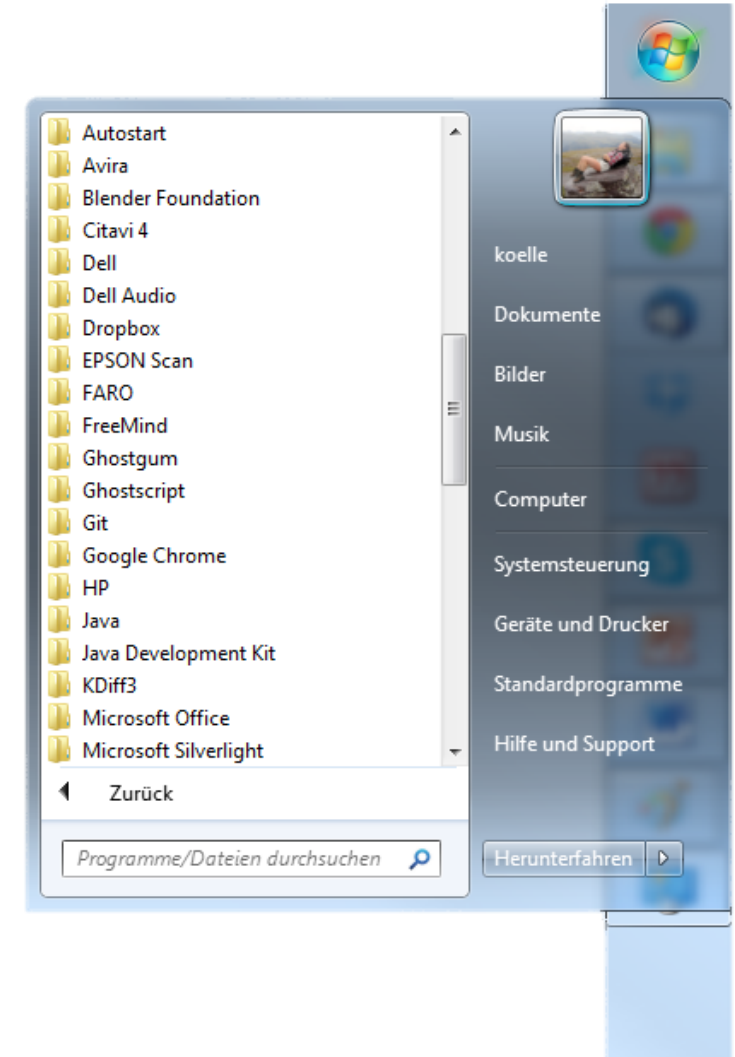


Human Interface Devices

Human Input: Selection Performance

→ Hick's Law Examples

- Used to decide how menus are arranged
- Searching for a certain program (name) in an alphabetically sorted list, search is logarithmic
- Hicks' Law does apply

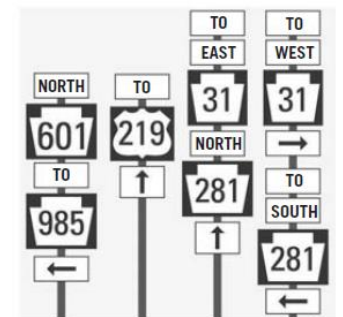
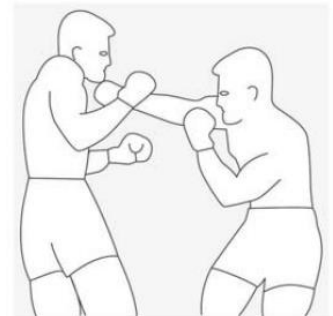
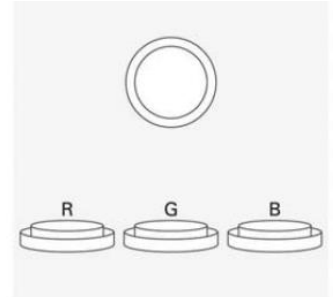


Human Interface Devices

Human Input: Selection Performance

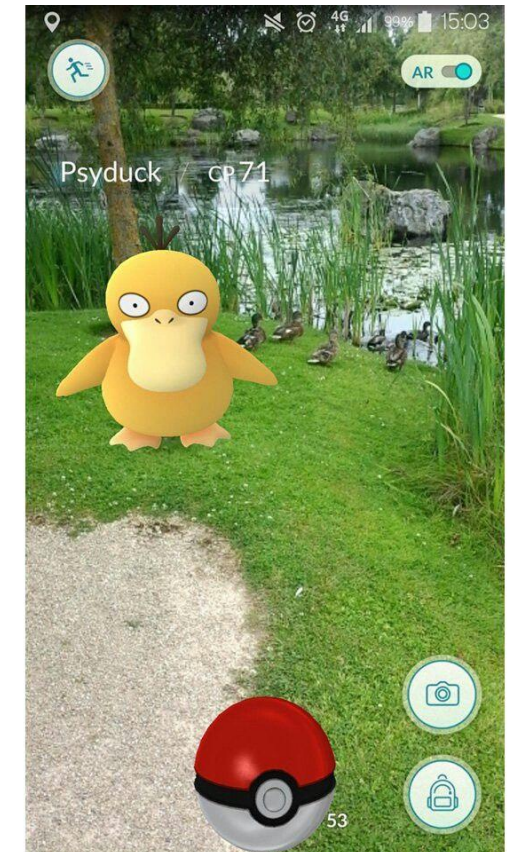
→ Hick's Law: Interesting facts

- **Simple tasks:** The time needed to select a (colored) button depending on the color of a light increases with the number of options
- **Martials arts:** The time needed to block an attack increases with the number of blocking techniques a fighter is capable of
- **Traffic signs:** The time needed to decide for a turn increases with the number of traffic signs
- **Braking:** the time needed to apply the brake pedal in case of an unexpected event (i.e., animal on the street) increases with the number of options to prevent the crash



Human Interface Devices

Human Input: Task Analysis



Human Interface Devices

Human Input: Task Analysis

→ Hierarchical Task Analysis

- **What** is the analysis for (i.e., assist system design, training and learning, etc.)?
- Define **goals**
- Obtain necessary **data** (i.e., user observation, literature review, etc.)
- Draft a hierarchical **diagram** (based on obtained data)
- **Evaluate** the decomposition (amount of detail, abstraction; with stakeholders)
- **Identify** important operations (subtasks that fit the necessary detail)
- **Design based on “selecting/supporting touchpoints in space and time”**

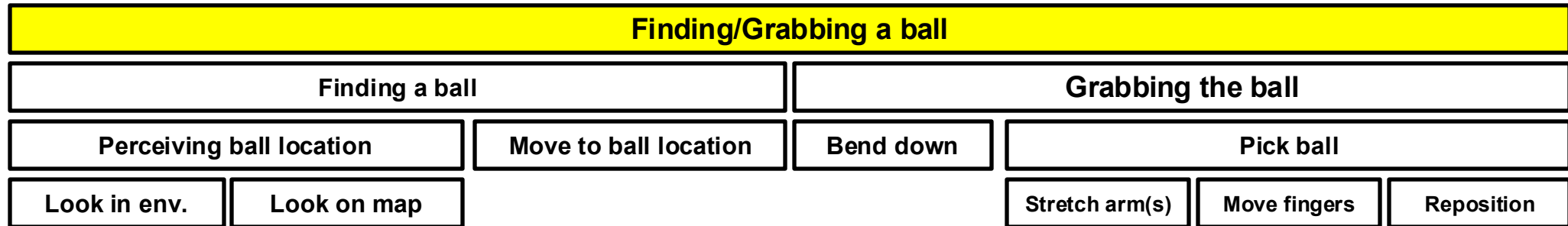
Human Interface Devices

Human Input: Task Analysis – Shooting a Volleyball



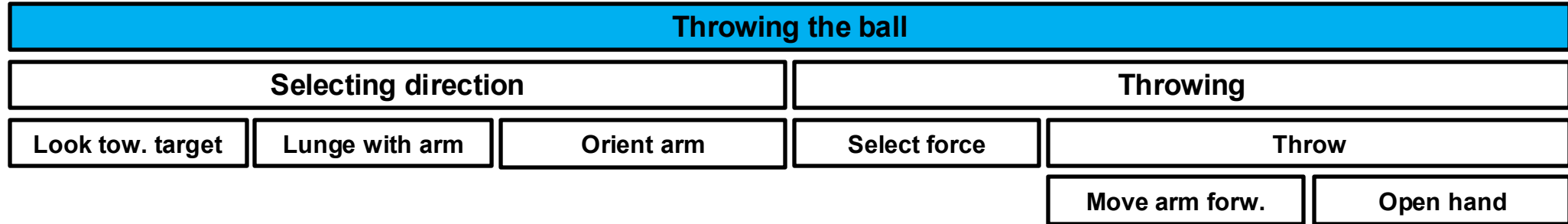
Human Interface Devices

Human Input: Task Analysis – Shooting a Volleyball



Human Interface Devices

Human Input: Task Analysis – Shooting a Volleyball



Human Interface Devices

Human Input: Task Analysis – Shooting a Volleyball



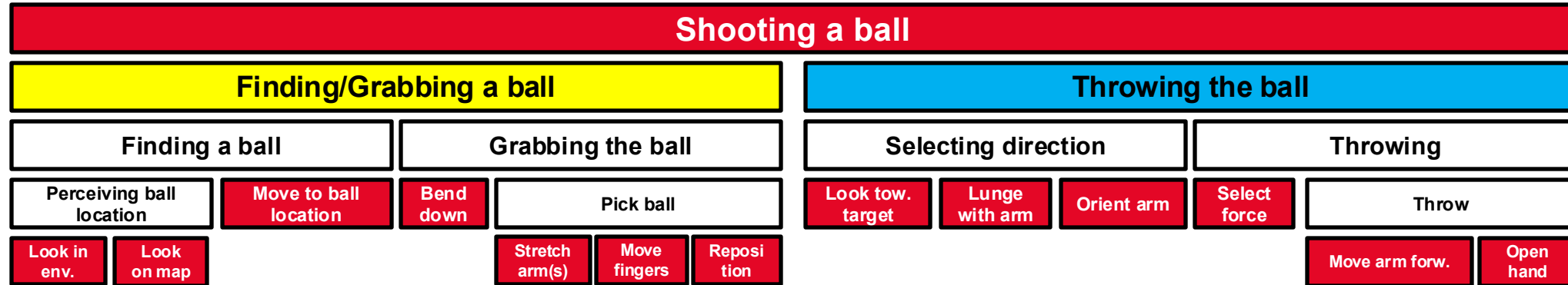
Design

What are the users' tasks?

Human Interface Devices

Human Input: Task Analysis – Shooting a Volleyball

Design
Virtual Reality



Human Interface Devices

Human Input: Task Analysis – Shooting a Volleyball

Design

Keyboard/Mouse Game



The user can simply click and finds a ball in their hands

The user aims with a mouse

The user chooses the strength by click duration

Game Engine Concepts



UNIVERSITY
OF APPLIED SCIENCES
UPPER AUSTRIA

FH Oberösterreich

University of Applied Sciences Upper Austria

Digital Media Department

Media Technology & Design

Digital Arts

Interactive Media



Dr. Andreas Riegler

Professor of Interactive Systems

Digital Media Department

E Andreas.Riegler@fh-hagenberg.at